

ICTTEN202 Use hand tools

General Hand Tool Safety



INFORMATION

The procedures and recommendations included in this resource are for *general information only*. Hand tools differ in names, design and functions between manufacturers and models.

Always consult manufacturer's information. Hand tools can be dangerous when used incorrectly.

The author would like to thank the listed tool manufacturers for the use of their images and information where applicable.

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Referenced text

The following documents are referenced in this resource:

- Gearwrench Product Catalogue
- Band-it-strap Tool Operating Instructions
- Wiha Professional Insulated Tools Catalogue
- Knipex General Catalogue
- Crescent Tool Catalogue
- Nicholson Product Catalogue
- Nicholson - The guide to files and filing
- H.K.Porter Tool Catalogue
- Sutton Tool Catalogue Threading
- Ridgid Tool Catalogue
- Plumb Tool Catalogue
- Dawn Product Catalogue
- Proto Pullers and Pullers Sets
- Utica Torque Products
- Wiss Product Catalogue

It is recommended to use the following two general tool catalogues for reference:

- Blackwoods E-Catalogue
- United Tools Product Catalogue

Application

This unit of competency defines the skills and knowledge required to use a range of hand tools for a variety of general engineering applications.

This unit should not be selected if the hand tool is dedicated to a single operation or machine or when a machine-specific/customised tool is used.

Learning outcomes

Using this *Learning Guide* you will develop an understanding of:

- the basic science of hand tool use
- the types of hand tools used by engineering tradespersons
- how to identify particular hand tools
- how to select the correct hand tool for the task
- the safety requirements when using hand tools
- how to correctly use the hand tool
- how to sharpen relevant hand tools
- to identify unsafe or faulty hand tools before, during or after use
- operational maintenance of hand tools
- correct storage of hand tools

Did you know?



A hand tool is a device for performing work using the hands. Humans have been using hand tools for millions of years and one of the first known hand tools, was a flint that was split and chipped to form a sharp edge. This sharp edge was then used to shape bone into even sharper tools.

It is not just humans who use hand tools to their advantage though. Chimpanzees strip twigs to form a tool and then plunge the tool into an anthill to catch the termites which cling to the end of it.



Hand tools used in engineering

Engineering tradespersons use a comprehensive range of hand tools for a wide variety of tasks. A hand tool is said to be an '**extension of the hand**'. Hand tools are used to:

- fabricate components to specifications e.g. shape, size, finish, tolerance, etc.
- adjust and maintain equipment e.g. tension, torque, etc.
- install equipment *and*
- fault find etc.

Hand tools can be broadly grouped as:

- fastening**
- gripping** and **clamping**
- cutting**
- impact** *and*
- measuring** and **marking** (these are included in *MEM12023A Perform engineering measurements*)

Typical hand tools used by an engineering tradesperson include the following:

- spanners and wrenches
- screwdrivers
- pliers and cutters
- files and saws *and*
- hammers and chisels etc.

Each hand tool has a specific purpose, design and proper use. Always abide by these when using hand tools. It is important to carefully follow manufacturer's instructions and recommendations.



Figure 1 Tool manufacturers (Courtesy of manufacturers)

General hand tool safety

Hand tools can be dangerous if used incorrectly. In this Learning Guide, specific precautions applicable to each tool are included with the tool description, however, always follow these *general hand tool safety tips*:

- Never use a tool that is not designed for the task. Decide on the right tool for the job before beginning. This includes the type, size and design of the tool

- Use quality made and ergonomically designed tools. This will prevent costly replacement and injuries. Tools can last a lifetime if maintained correctly
- Follow manufacturers' recommendations. There is usually a wealth of information in manufacturer catalogues, instruction or operation manuals and data sheets
- Wear appropriate personal protective equipment (PPE). The PPE required will depend on the work site and the tool being used.
- Gloves must only be close fitting. Contact dermatitis can occur from prolonged use; barrier cream can prevent allergic reactions
- Inspect tools before and after use for defects. If they are damaged in any way, place them out of service
- Maintain tools and store correctly – this includes keeping them in a toolbox or similar
- Keep your work environment clean and tidy – employ proper housekeeping procedures before, during and after the work activity
- Keep cutting tools sharp and handle the sharp edges with care
- Work on isolated equipment or machinery. Follow workplace isolation, locking and tagging procedures before using tools
- Concentrate on the job you are doing
- Always secure work if possible – use a vice, pliers or clamps etc.
- Make sure that using the tool will not affect other people
- Hand tools can generate ignition sources – never use in a hazardous area without area de-classification
- If working from a ladder, tools should be raised and lowered using a hand line
- Use tools that are designed to keep the wrist straight – avoid using any tool with a bent wrist

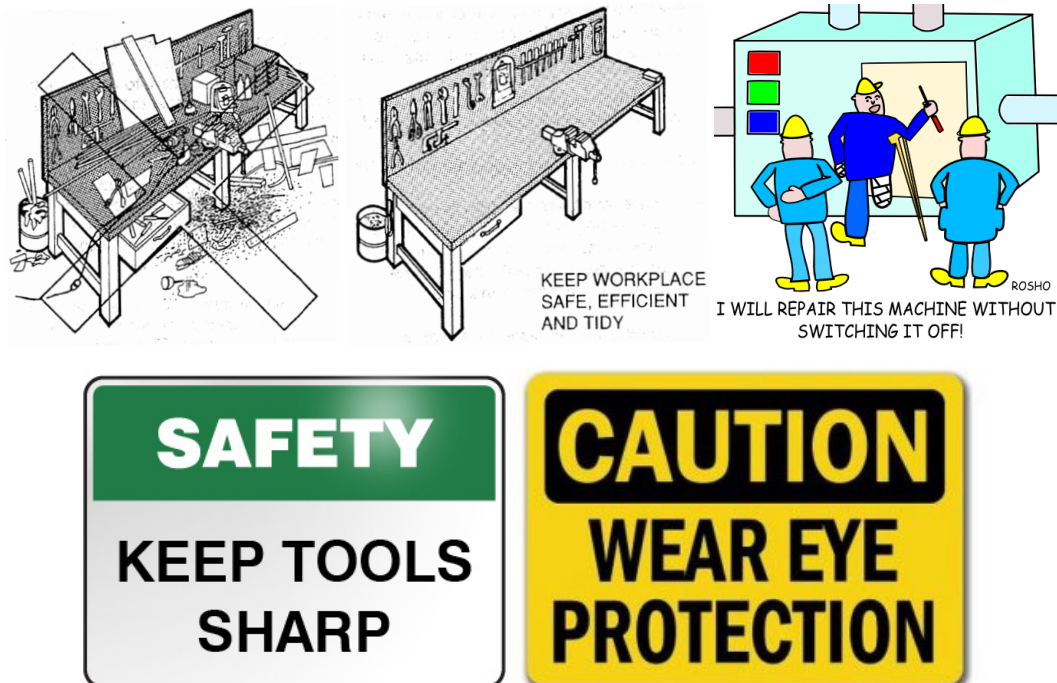


Figure 2 Always follow safe working procedures

The science of hand tools

Mechanical advantage

Most tools make use of **mechanical advantage**, which is 'a measure of the force amplification achieved by using a tool or other mechanical device'. The tool should use the applied force and movement provided by the operator, to amplify the output force. Mathematically, the mechanical advantage is 'the ratio of the output force produced by the tool which acts on a load to the applied input force'. Three examples of mechanical advantage are shown in Figure 3.

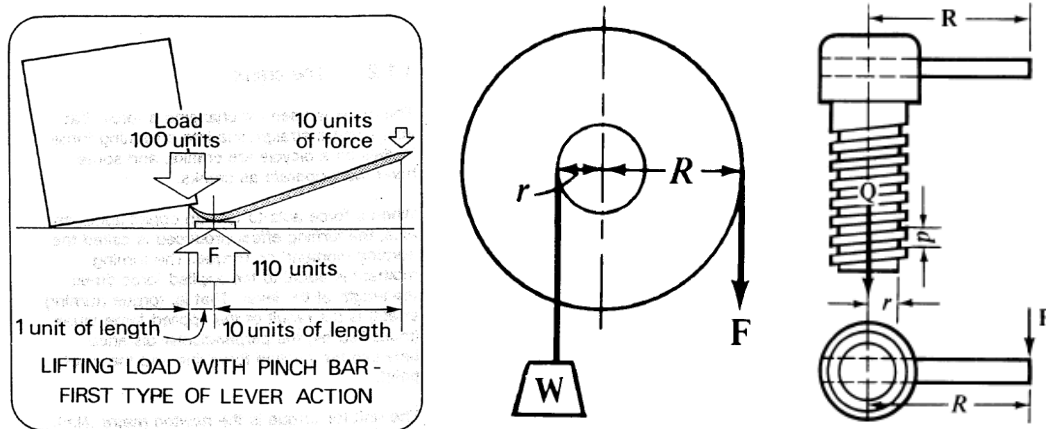


Figure 3 Mechanical advantage

'Give me a lever long enough and a fulcrum on which to place it and I shall move the world'

Archimedes

When you use hand tools, you are using many scientific principles to make the tool do its work. Following is a brief description of these common principles.

Leverage

A **lever** is able to produce a large force at one end by applying a much smaller effort at the other end. It is used for:

- pushing
- pulling
- lifting
- lowering etc.

A lever has a **fulcrum** (a point of support). The closer the load is to the fulcrum and the longer the lever is to apply the effort, the easier it is to shift a big load. Conversely, the closer the load is to the fulcrum, the smaller the distance the load can be moved. This can be seen in Figure 4 (L).

Leverage is a principle that is related to using a lever. It is related to the length of the lever (the tool) and the applied force. More leverage can be applied to a load with a longer tool, than a shorter one. This can be seen in Figure 4 (R).

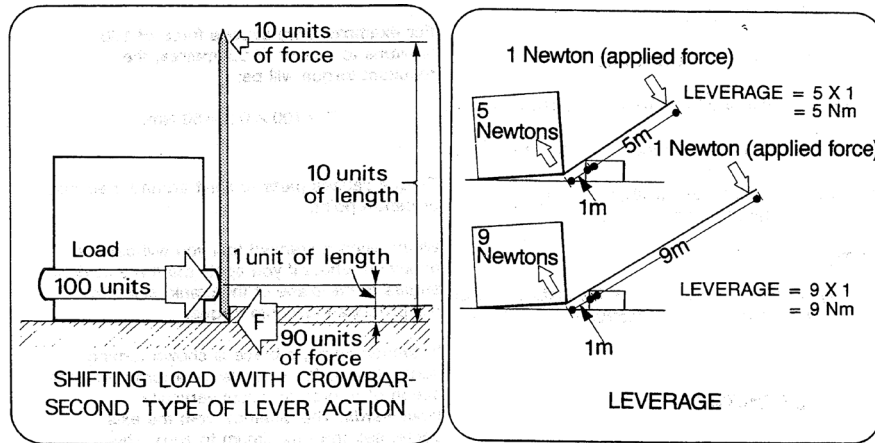


Figure 4 Leverage to a load

Torque

A **crank** is capable of producing a rotating force. Some hand tools operate as cranks. This rotating force (**torque**) is equal 'to the applied force multiplied by the length of the lever'. Torque is measured in newton-metres (Nm).

When using a tool to produce torque, maximum twisting action will be achieved if you apply force at right angles to the plane of the crank compared to perpendicular. The difference is shown in Figure 5.

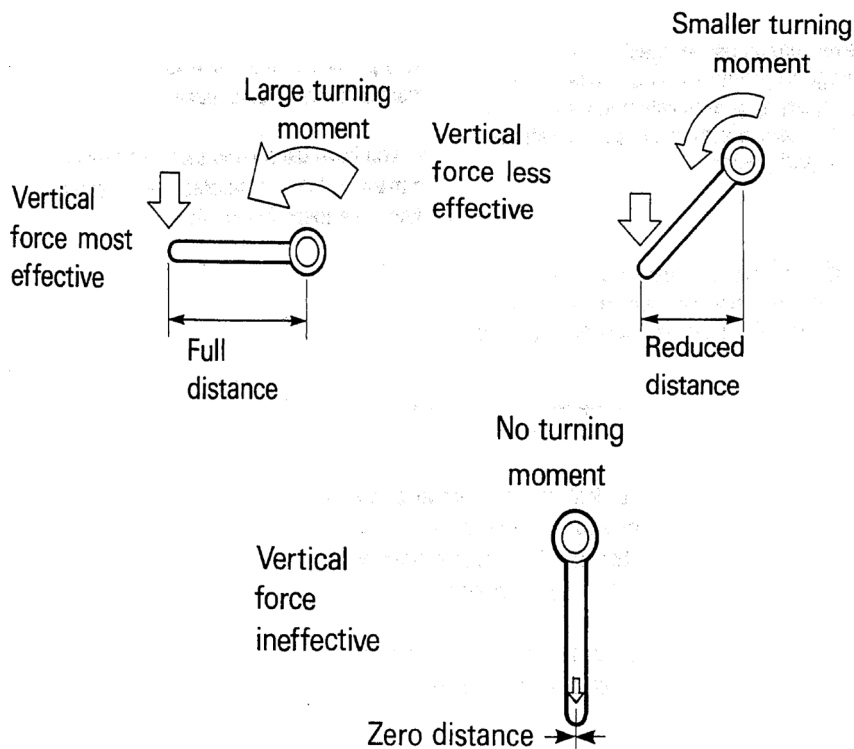


Figure 5 Production of torque using the crank

Force

Applied **force** is the quantity of push or pull given to the tool by the operator. It is the amount of torque applied to a point and is measured in newtons (N). Figure 6 shows force being applied to a spanner.

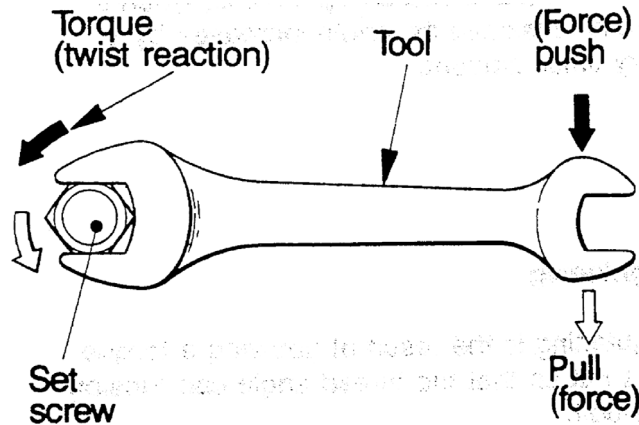


Figure 6 Force applied by a spanner

Pressure

When force is applied to a small area, it produces high **pressure**. When the same force is applied to a larger area, lower pressure is produced. This is shown in Figure 7. Hand tools such as chisels and saws apply concentrated pressure to a small area.

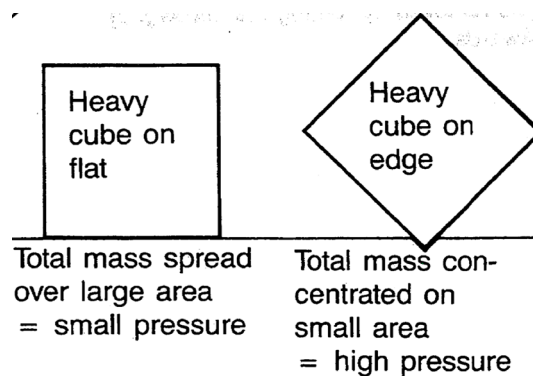


Figure 7 Pressure exerted on an area

Tension

Applying **tension** means to stretch. When tension is applied to a bolt, its length *increases*. **Tightening** is the result of applying torque to a nut so that the bolt is tensioned. However, this applied force will also twist the bolt. Tensioning is shown in Figure 8.

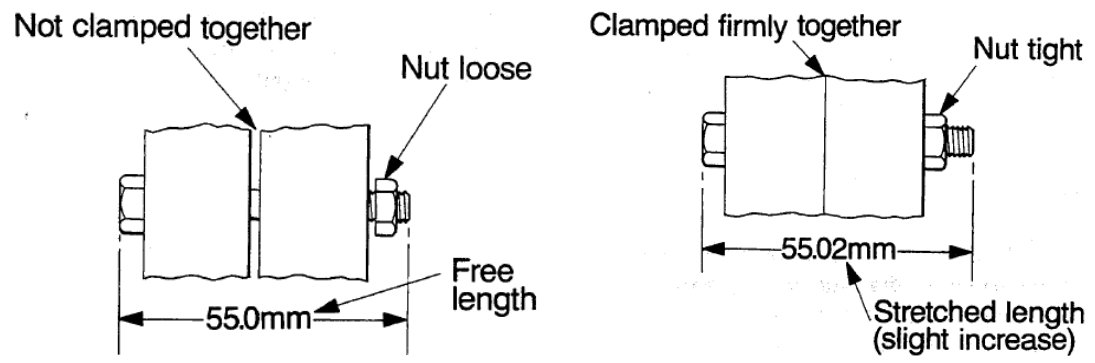


Figure 8 Tensioning a bolt